

Project Design Phase
Proposed Solution Template

Date	13 February 2026
Team ID	LTVIP2026TMIDS82993
Project Name	HouseHunt: Finding Your Perfect Rental Home
Maximum Marks	2 Marks

Proposed Solution Template:

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Finding rental houses is often difficult due to scattered listings, broker dependency, lack of transparency, and risk of fraudulent advertisements. Property owners also struggle to connect directly with genuine tenants without paying high commissions. There is a need for a centralized, secure, and role-based digital platform to manage rental properties efficiently.
2.	Idea / Solution description	HouseHunt is a MERN stack-based web application that connects property owners and renters through a secure and centralized platform. The system provides user registration and login with JWT authentication, role-based dashboards (Admin, Owner, Renter), property listing and management features, and booking functionality. Owners can add and manage properties, renters can browse and book properties, and admins can monitor system activities.
3.	Novelty / Uniqueness	The system integrates role-based authentication, secure property management, and booking tracking within a structured 3-tier MERN architecture. Unlike traditional broker-based systems, HouseHunt eliminates middlemen by enabling direct interaction between owners and renters. The modular backend design and scalable structure make it adaptable for future enhancements like payment gateways and cloud deployment.
4.	Social Impact / Customer Satisfaction	HouseHunt reduces dependency on brokers and minimizes rental fraud. It improves transparency, enhances trust between users, and simplifies the house searching process. Renters get verified listings and owners get direct tenant communication, leading to higher customer satisfaction and reduced financial exploitation.

5.	Business Model (Revenue Model)	The platform can generate revenue through: • Commission per successful booking • Featured property listing fees • Subscription plans for premium owners • Advertisement placement • Future integration of payment gateway transaction charges
6.	Scalability of the Solution	The application follows a 3-tier architecture (Frontend–Backend–Database), making it scalable. It can be deployed to cloud platforms such as AWS or Render. The modular code structure allows easy addition of new features like payment integration, mobile app development, and microservices expansion. The system supports horizontal scaling using load balancers and cloud database services.